

Sahir Karani

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EXPERIENCE

Amazon Prime Video, New York, NY

February 2021 – Present

Senior Software Engineer

- Drove technical strategy for a 25+ engineer cross-org initiative to design and deliver a long-term machine learning platform strategy, supporting 100+ models (DL, LLM) across 3 orgs and reducing training and deployment time by 73%.
- Developed ML driven ranking and personalization pipelines, integrating real-time inference to boost click through rates by 18% and increase average stream time by 78 min through model tuning and A/B experimentation.
- Designed and implemented 7 Tier-1/Tier-2 distributed personalization and recommendation services handling 300K+ TPS with 99.99% availability and < 300 ms p99 latency via dependency tiering, graceful degradation, regional failover, adaptive throttling, and automated recovery.
- Defined org-wide standards for generative AI adoption in software development, authoring best practices and code examples for 450+ engineers, increasing developer efficiency by 30% and reducing code review cycle times by 40%.
- Architected system design for multi-region microservices across 4 AWS regions, enabling 3x on-demand scaling during major live events and cutting operational costs by ~20% via caching and auto-scaling serving 200M+ customers.
- Mentored 5+ engineers through design reviews, code reviews, and technical guidance, contributing to 4 promotions.

Goldman Sachs, New York, NY

July 2018 – February 2021

Software Engineer - Quantitative Analyst

- Developed a full-stack risk management application to support upsizing and downsizing of hedged swaps across \$530M in alternative investments, reducing risk exposure decision time by 40%.
- Led architectural blueprinting of total returns swaps order management system supporting \$1T+ in assets under management (AUM), defining scalable workflows for trade lifecycle and controls.
- Developed a configurable rules engine and UI for credit agreement management, supporting \$18B in exposure and reducing manual review and configuration errors by 30%.

J.P. Morgan Chase & Co, New York, NY

June 2017 – July 2018

Software Engineer Analyst

- Researched and evaluated Attribute-Based Access Control (ABAC) to modernize entitlement management, influencing a shift from role-based access across 3 teams and improving security flexibility via JPMC Security & Architecture Council.
- Designed and implemented a centralized Policy Decision Point (PDP) microservice using Python, XACML, and Flask, enforcing consistent authorization across 8 services and supporting security best practices.
- Led Agile Scrum execution and stakeholder management for the project, improving team velocity by 30%.

Deloitte, New York, NY

June 2015 – June 2017

Networks Engineer Analyst

- Developed Python-based network automation scripts for LAN, WAN, Wireless, and IP management solutions reducing manual configuration effort by 90%+ and improving deployment across 80+ US office sites.
- Troubleshoot and resolved network outages and performance issues, performing fault isolation and recovery to restore service within 2.5 min from detection and failover to other network providers automatically.

EDUCATION

New York University, Tandon School of Engineering, New York, NY

B.S. in Computer Science, Cyber Security Minor | Honors: *magna cum laude* (GPA: 3.8/4.0)

AWARDS AND ACTIVITIES

- **Hackathons:** Prime Video PVPD Hackathon - Winner, Prime Video CDEX Hackathon - Winner, Goldman Sachs Datathon - Top 5, Goldman Sachs Global Hackathon - Winner, HackNYU - Best Android Hack, WearHacks - Best Software Hack
- **Athletics:** Marathon (2x), Half Marathon (9x), Summited Mt. Kilimanjaro, Spartan Race (6x), Bike Tour (5x)
- **Volunteering:** Aga Khan Economic Planning Board, The Opportunity Network

TECHNICAL SKILLS AND CERTIFICATIONS

- **Development:** Python, Java, Kotlin, C++, Distributed Systems, Microservices, System Design, REST APIs
- **ML/AI:** Deep Learning, LLMs, Neural Networks, Reinforcement Learning, Real-time Inference, PyTorch, NumPy
- **Tools:** AWS (EC2, S3, DynamoDB, SageMaker, Lambda, CloudWatch, Kinesis, Bedrock), Kubernetes, Docker, CI/CD
- **Certifications:** Stanford Machine Learning Specialization, FAA Airframe License, FCC Operator License